Motivation Letter for Research Stay

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Research stay at the Mobile Perception Systems Lab, Eindhoven University of Technology

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I am writing to apply for a three-month research stay at the Mobile Perception Systems Lab, Eindhoven University of Technology, between February and June 2027, funded by the Slovak National Scholarship Programme. I am a final-year Master's student at FIIT STU Bratislava, working on trajectory prediction for autonomous driving.

I came across the Mobile Perception Systems Lab while looking for research groups whose work connects to my Master's thesis, and found that the overlap was closer than I expected. My thesis is on perception for self-driving vehicles: teaching a system to detect the cars, pedestrians and cyclists around it from sensor data, to keep track of each one as it moves, and to anticipate where each is going. The lab describes its long-term goal in almost those terms, as building systems that do not only detect what is present at a given instant but also reason about how a scene will develop. That is the problem I have spent the past year on, which I have approached through trajectory prediction.

One of the questions that held my attention while working on trajectory prediction was how the components should pass information to one another. Conventional systems are built as a chain of separate modules for detection, tracking and prediction, each passing the next a compressed summary of what it found. Every hand-off discards detail that the next stage needed. The line of research I follow argues instead for carrying a richer internal record of each object through the whole system, so that nothing is thrown away between stages.

Reading further into the group's recent direction led me to a class of driving models with which it has begun to work, in which the system writes a short explanation of what it intends to do and why, in ordinary language, before computing the maneuver. The explanation is not commentary added afterwards: the steering and braking are derived from it. Looking closely at the architecture, I noticed that the model does not maintain an internal record of the objects it has actually perceived. It refers to them only in words. Nothing in the system requires the pedestrian it mentions to exist, and its authors acknowledge that it sometimes describes objects that are not there. That is the same problem I had been working on, one level higher in the system, and the kind of representation my thesis is built on is close to the one that is missing.

The project the group is building around these models is what makes the stay particularly valuable to me. It spans the full driving stack, from sensing through reasoning and planning to deployment on the vehicle itself and in simulation. My own work so far has covered one segment of that pipeline. Contributing to a system where the whole stack is being assembled at once, and against a real vehicle rather than only a benchmark dataset, would be a great opportunity to broaden my understanding of autonomous driving as a whole.

My thesis will be mostly completed this semester, so extending it with work that requires a real vehicle and a full driving stack is the natural next step. I have also successfully completed the summer semester of 2025/26 on an Erasmus exchange at FIT CTU Prague, and so have experience with studies at a foreign institution. As I intend to continue to a doctorate in artificial intelligence, I see a semester in an established group as an enriching step in my academic development.